$a^{[1]} = x + \varepsilon_1$

$a^{[2]} = x + \varepsilon_1 + \varepsilon_2 = a^{[1]} + \varepsilon_2$ → getting cascade.

after 500 layer

$a^{[500]} = a^{[499]} + \varepsilon_{500}$

Solution:-

Relu is an activation function.

$a > 0$

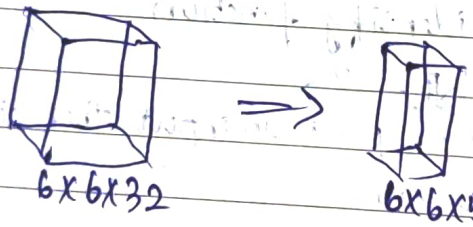
$$a^{[l+2]} = g\left(w^{[l+2]} a^{[l+1]} + b^{[l+2]} + a^{[l]}\right)$$

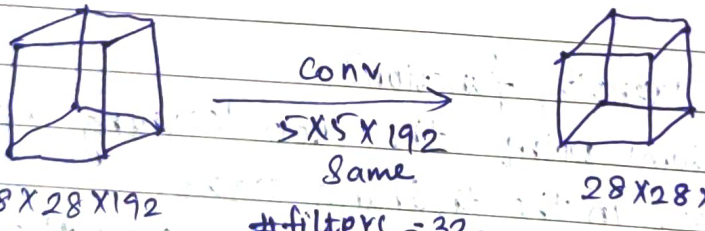
since $w = 0 \quad b = 0$

$$= g(a^{[l]})$$

$$= a^{[l]}$$

1 x 1 convolution.





Total no. of computations / Multiplications?

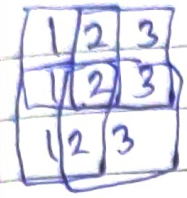
1 pixel = 5x5x192

28x28x32 = 120,422,400.

Just classwork

1 2 3
4 5 6
9 8 9

3X3 - layer
2X2 - 16.



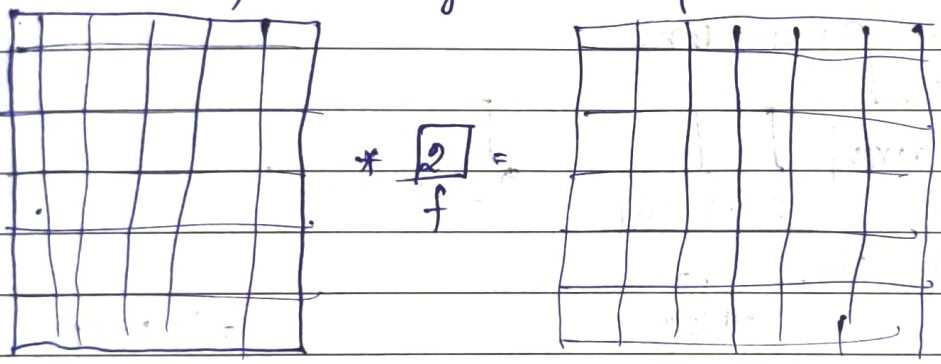
1 0 =
0 1

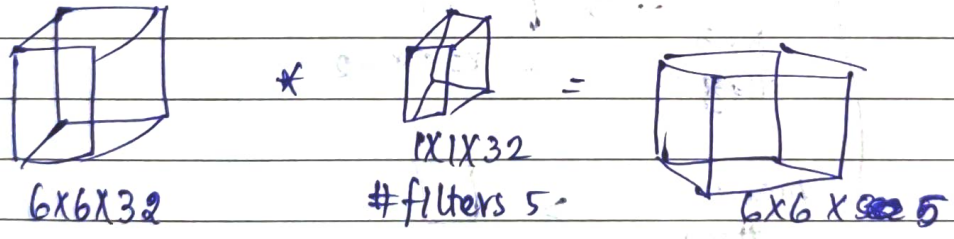
28X28X192 → 5X5X192

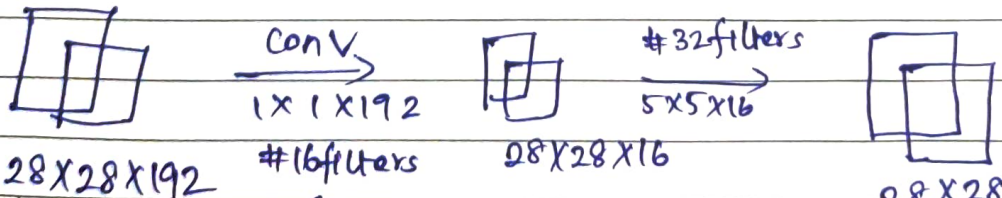
24X24X192 = X32
↓ ↳ 3538944
110592

24X24X32 = 18432

1X1 convolution :- Network in a Network.
1) Reducing the no. of channels
2) Reducing the computational complexity.







$28 \times 28 \times 32 \times 5 \times 5 \times 16 = 10035200$

$+ \; 28 \times 28 \times 16 \times 192 = 2408448$

$12,443,648$

120422400

12443648

107978752

Inception Network :-

17.08.2026

Lecture 20 :- Inception network - GoogleNet

- Transfer learning

check the parameters.

